CloudOps: Continuous Operations Using Ansible and Jenkins

CloudOps, as the name suggests, comprehensively covers all the tasks required to run a set of cloud based business applications. Organisations are migrating their IoT systems to the cloud to enhance scalability while optimising performance and capacity. In this article, we highlight how Ansible and Jenkins can be used for CloudOps.

When the infrastructure that supports technology changes structurally, it is essential to understand this change in order to make strategic decisions about the future of a business or project. Cloud computing responds to this need by accelerating the development and scalability of technologies such as artificial intelligence, the Internet of Things and machine learning, optimising processes and assisting in strategic decision making. Regardless of which platform is used or the location of the infrastructure, CloudOps provides organisations with proper resource management. It uses DevOps principles and IT operations applied to a cloud based architecture to speed up business processes.

CloudOps relies on continuous operations and this approach has been taken from DevOps. It is similar to DevOps but used for the cloud platform, and has different flavours like native operations, third party operations and agnostic operations. CloudOps differs from DevOps in that it concentrates more on task automation and cost optimisation factors for the *cloud* infrastructure, platform and application services.

Ansible reference architecture

Ansible is an important open source automation tool or platform used for diverse IT tasks like configuration management, application deployment, intra-service orchestration and provisioning. Automation is crucial for systems administrators as it simplifies tasks so that attention can be focused on other important activities.

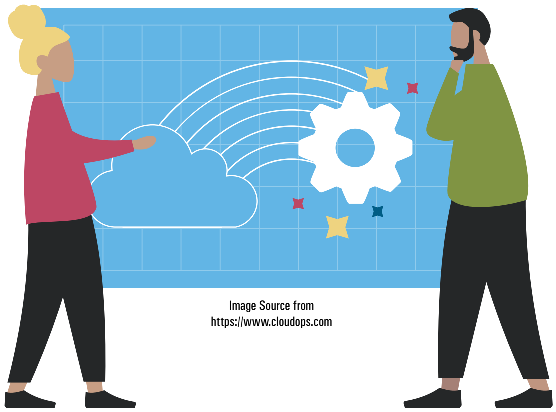

Image Source from https://www.cloudops.com

Ansible, being an open source tool, is very easy to understand and doesn't use any sort of third-party tool or complex mechanism for security. It uses playbooks to perform end-to-end automation of varied components. Playbooks make use of YAML (yet another markup language) to manage all files. Ansible is available for both single- and multi-tier systems and infrastructures.

Here is an example of how playbooks are used. A company releases software, and to make this software functional it is important to have the latest version of the WebLogic server on all machines. It is cumbersome for enterprise administrators to manually check that this has been done across all the machines. The best alternative is to install an Ansible playbook written in user-friendly syntax and YAML, and then run it from the central machine. All nodes (clients) connected to the main machine get updates and make all the necessary changes required to run the software. Figure 1 highlights the reference architecture of Ansible.

The terms used in this architecture are briefly explained below.

User: This refers to the end user who creates and defines the Ansible playbook, which has a direct connection with the Ansible automation engine.

Ansible playbook*:* Playbooks comprise code written in YAML, which describes the tasks and their execution via Ansible. These tasks can be synchronously and asynchronously run via playbooks. Playbooks are the heart of Ansible as they not only declare configurations, but can orchestrate the steps of any manual tasks for execution at the same time or at different times on all connected machines. They interact with the Ansible automation engine and configuration management database.